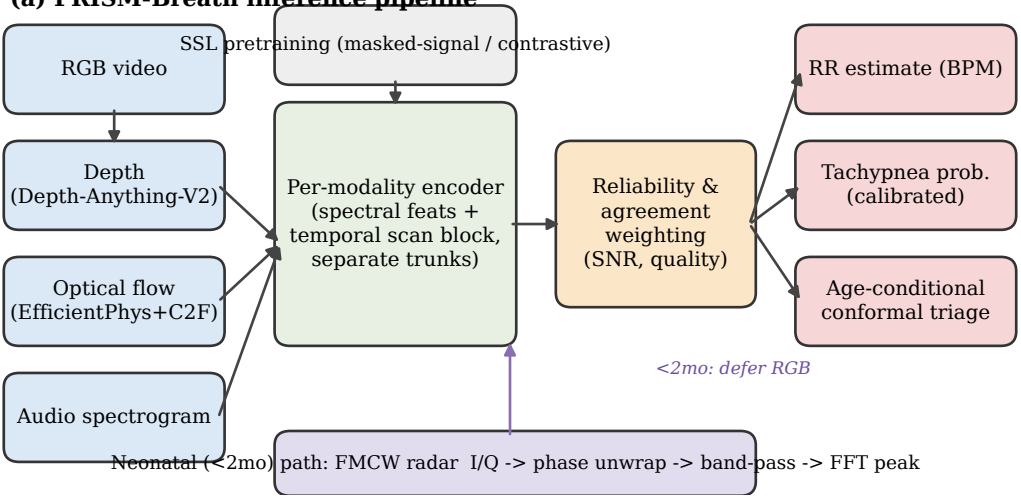
(a) PRISM-Breath inference pipeline
SSL pretraining (masked-signal / contrastive)
RGB video
Depth (Depth-Anything-V2)
Optical flow (EfficientPhys+C2F)
Audio spectrogram
Per-modality encoder (spectral feats + temporal scan block, separate trunks)
Reliability & agreement weighting (SNR, quality)
RR estimate (BPM)
Tachypnea prob. (calibrated)
Age-conditional conformal triage
<2mo: defer RGB
Neonatal (<2mo) path: FMCW radar I/Q -> phase unwrap -> band-pass -> FFT peak